Ethan Kim

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EDUCATION

Harvey Mudd College — Claremont, CA

Bachelor of Science in Engineering — *Expected May 2029*

- Relevant Coursework: Engineering Design & Manufacturing; Engineering Systems; Materials Engineering; Probability & Statistics
- Upcoming Coursework: Continuum Mechanics; Mechanical Design; Dynamics of Rigid Bodies; Experimental Engineering

EXPERIENCE

Cenergy Power — Utility-Scale Solar Development — Aliso Viejo, CA

Engineering Intern — *June 2026 – August 2026*

- Automated parcel screening for utility-scale solar development across Virginia and Illinois, filtering candidate sites by proximity to substations (<5 mi) and distribution feeders (<1 mi) which reduced county-level research from 16 hours of manual work to 15 minutes.
- Developed a system overlaying regulatory and environmental constraint layers (FEMA flood zones, prime farmland, wetlands) onto AutoCAD parcel drawings to give designers buildable-area context before site layout.
- Built an end-to-end landowner outreach pipeline that aggregates parcel and ownership records to generate print-ready mailer packets with preliminary site designs, deployed across all target counties.

DWK Corporation — Ontario, CA

Marketing & Forecasting Analyst — *May 2020 – May 2024*

- Diagnosed underperforming Amazon listings through keyword research and bid optimization to increase sales by 5%; built Excel forecasting models for inventory stocking and demand projections.

PROJECTS

Harvey Mudd College Robotic Submarine Team — Claremont, CA

Lead Mechanical Designer — *September 2025 – Present*

- Modeled structural and housing components in SolidWorks for "Crush," the team's autonomous underwater vehicle, from concept through manufacturable part drawings, simulating stress and strain on the casing to confirm margin before fabrication.
- Select materials and seal geometry for sustained submerged operation, weighing strength-to-weight against corrosion resistance.
- Made design drawings, process routings, and machined parts for "Crush" utilizing lathes and mills.

Harvey Mudd College Formula SAE Racing Team — Claremont, CA

Mechanical Engineer — *September 2025 – Present*

- Contribute to chassis fabrication and integration for the team's competition vehicle.
- Work with a two-year-old program where members work across sub-teams rather than specializing, from fabrication to assembly.

Wheeev — iOS App — Irvine, CA

Solo Developer — *July 2026 - Present*

- Designed, built, and shipped a live App Store iOS app for the car and motorcycle community to discover local meets, message other riders, and share real-time location. Built with React Native (Expo) and TypeScript; Supabase (PostgreSQL + PostGIS).

National Association of Rocketry — Annual Meet Championship — Pueblo, CO

Silver Medalist — *June 2024 – August 2024*

- Placed 2nd nationally in Helicopter Duration by tuning mass across builds to align apogee with a fixed motor ejection delay
- Designed, built, and flight-tested 5 custom carbon fiber prototypes over a 3-month period, isolating one variable per build to identify what drove flight duration and converge on a final configuration.

Autonomous Trash Rover — Independent Capstone Project — Irvine, CA

Lead Engineer — *May 2025 – June 2025*

- Built an autonomous ground rover in Python integrating computer vision, touch sensors, and motor control to detect and collect litter; field-tested over a 4-week cycle, retrieving 50 pieces of debris

SKILLS & INTERESTS

- **Mechanical Design:** SolidWorks (parts, assemblies, drawings), AutoCAD (layered site drawings), material selection, design iteration
- **Manufacturing**: Manual mill and lathe, 3D printing, composite layup, hand tools and assembly
- **Programming & Data:** Python (workflow automation, data pipelines), TypeScript / React Native, SQL (PostgreSQL / PostGIS), Excel
- **Interests:** Motorcycles (ride and maintain a Ninja 300), Formula 1, MotoGP, model rocketry, drone racing, coaching tennis

ACTIVITIES & LEADERSHIP

Claremont-Mudd-Scripps Varsity Tennis (NCAA Division III) — Claremont, CA

Student-Athlete — *August 2025 – November 2026*

- Competed and contributed 12 hours per week to a program ranked #2 nationally in Division III.
- Won the 2026 Division III National Indoor Championships and runner up in the NCAA Championships in Chattanooga.